

Community Blood Donor Finder

100% Working-Model Prototype Documentation

Purpose

A browser-based educational coordination prototype for donor registration, emergency blood requests, potential donor matching, availability/location prioritization, donor-alert simulation, and an admin dashboard. It runs locally without a backend.

Important scope

The prototype is a support/coordination platform. It must not make medical decisions. Medical screening, donor eligibility and transfusion compatibility remain with qualified healthcare professionals or blood-bank staff.

Technology

HTML5 + CSS3 + vanilla JavaScript. No framework, database, server, API key, or external dependency is required.

Prototype workflow

Step	User action	System response
1	Open HTML file	Home dashboard loads with donor/request counters.
2	Register donor	Name, blood group, location and availability are stored in browser memory.
3	Create emergency request	Blood group, location and units are captured.
4	Match	Available donors with the requested blood group are identified.
5	Prioritize/display	Potential matches show blood group, availability and approximate location.
6	Alert simulation	A donor-alert message is generated for each potential match.
7	Admin review	Dashboard shows donor count, request count and donor records.
8	Hospital coordination	Final eligibility/compatibility and donation process are handled externally by professionals.

Testing section

ID	Test	Input	Expected result
T01	Page load	Open HTML in browser	Home page loads; counters and navigation are visible.
T02	Donor registration	Name=Test Donor, blood=A+, location=Coimbatore, Available	Record added; donor count increases; record appears in Admin.
T03	Empty name validation	Leave donor name blank	Browser alert asks for donor name; no record is added.
T04	Positive match	Request O+	Available O+ donor(s) appear under Nearby Potential Matches.
T05	Busy donor exclusion	Request O+ while Rahul is Busy	Rahul is not shown as a potential match.
T06	No match	Request B+ with no available B+ donor	No-currently-available notice is shown; no donor alerts are generated.
T07	Alert simulation	Create request with a matching donor	Alert entry identifies the potential donor and request area.
T08	Admin sync	Register another donor	Admin donor count and donor table update immediately.
T09	Navigation	Click all six navigation buttons	Only the selected section is visible.
T10	Responsive view	Resize browser/mobile width	Layout reflows using responsive CSS.

Demo procedure

1. Save/open the supplied HTML file in Chrome, Edge, or another modern browser.
2. Click **Donor Registration**. Add a new donor, choose a blood group and set availability to Available. Submit.
3. Open **Emergency Request**. Select the same blood group and enter the hospital/request location and units.
4. Click **Create Request & Find Matches**. The matching donor(s) appear automatically.
5. Open **Donor Alerts** to show the simulated notification.
6. Open **Admin Dashboard** to show updated donor/request counts and donor records.
7. For a no-match demonstration, select a blood group for which no available donor exists; the system displays a no-match notice.

Complete HTML/CSS/JavaScript source

The following is the complete standalone source used by the working prototype. Save it exactly as **community_blood_donor_finder_working_prototype.html** and open it in a browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Community Blood Donor Finder</title>
<style>
*{box-sizing:border-box}
body{margin:0;font-family:Arial,sans-serif;background:#f5f7fb;color:#172033}
header{background:#9f1239;color:#fff;padding:18px 6%;display:flex;justify-content:space-between;align-items:center}
header h1{margin:0;font-size:24px}.badge{background:#fff;color:#9f1239;padding:7px 12px;border-radius:20px;font-weight:bold}
nav{background:#fff;padding:10px 6%;display:flex;gap:8px;flex-wrap:wrap;box-shadow:0 2px 8px #0001}
nav button,.primary{border:0;padding:11px 16px;border-radius:8px;cursor:pointer;background:#9f1239;color:#fff;font-weight:bold}
nav button{background:#eee;color:#333}
.container{max-width:1100px;margin:25px auto;padding:0 18px}
.hero{background:linear-gradient(135deg,#9f1239,#be123c);color:white;border-radius:18px;padding:30px}
.hero h2{font-size:34px;margin:0 0 10px 0}.hero p{max-width:720px;line-height:1.6}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(240px,1fr));gap:16px;margin-top:20px}
.card{background:#fff;border-radius:14px;padding:20px;box-shadow:0 4px 18px #000000d}
.card h3{margin-top:0}.stat{font-size:30px;font-weight:bold;color:#9f1239}
label{display:block;margin:10px 0 5px 0;font-weight:bold}
input,select,textarea{width:100%;padding:11px;border:1px solid #ccd3df;border-radius:8px}
textarea{min-height:90px;resize:vertical}
.formgrid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:12px}
table{width:100%;border-collapse:collapse;margin-top:15px}
th,td{padding:11px;border-bottom:1px solid #eee;text-align:left}th{background:#f1f5f9}
.status{padding:5px 9px;border-radius:15px;font-size:12px;font-weight:bold}
.available{background:#d4cfe7;color:#166534}.busy{background:#fee2e2;color:#991b1b}
.notice{padding:12px;border-radius:8px;background:#fff7ed;color:#9a3412;margin:12px 0}
.success{padding:12px;border-radius:8px;background:#ecfdf5;color:#166534;margin:12px 0}
.hidden{display:none}.match{border-left:5px solid #9f1239;padding:14px;background:#fff;margin:10px 0;border-radius:8px}
footer{text-align:center;padding:30px;color:#64748b}.small{font-size:13px;color:#64748b}
.danger{color:#b91c1c}
button:focus,input:focus,select:focus,textarea:focus{outline:2px solid #fb7185;outline-offset:1px}
@media(max-width:600px){header{gap:10px;align-items:flex-start;flex-direction:column}.hero h2{font-size:27px}}
</style>
</head>
<body>
<header>
<h1>■ Community Blood Donor Finder</h1>
<span class="badge">Working Prototype</span>
</header>

<nav>
<button onclick="show('home')">Home</button>
<button onclick="show('register')">Donor Registration</button>
<button onclick="show('request')">Emergency Request</button>
<button onclick="show('matches')">Nearby Matches</button>
<button onclick="show('alerts')">Donor Alerts</button>
<button onclick="show('admin')">Admin Dashboard</button>
</nav>

<div class="container">
<section id="home">
<div class="hero">
<h2>Find potential blood donors faster.</h2>
<p>A simple coordination prototype connecting blood requests with registered voluntary donors using blood group, availability and approximate location</p>
<button class="primary" onclick="show('request')">Create Emergency Request</button>
</div>
<div class="grid">
<div class="card"><div class="stat" id="donorCount">0</div><p>Registered donors</p></div>
<div class="card"><div class="stat" id="requestCount">0</div><p>Active requests</p></div>
<div class="card"><div class="stat">24/7</div><p>Coordination concept</p></div>
</div>
<div class="card" style="margin-top:20px">
<h3>How it works</h3>
<p><b>1.</b> Register donor → <b>2.</b> Create request → <b>3.</b> Match blood group → <b>4.</b> Check availability → <b>5.</b> Show potential matches</p>
</div>
<div class="notice">This is an educational coordination prototype. It does not make medical decisions, perform donor eligibility screening, or confirm matches</div>
</section>

<section id="register" class="hidden">
<div class="card">
```

```

<h2>Donor Registration</h2>
<div class="formgrid">
  <div><label>Name</label><input id="dName" placeholder="Donor name"></div>
  <div><label>Blood Group</label>
    <select id="dBlood"><option>A+</option><option>A-</option><option>B+</option><option>B-</option><option>AB+</option><option>AB-</option><option>O+</option><option>O-</option></select>
  </div>
  <div><label>Location</label><input id="dLocation" placeholder="Area / City"></div>
  <div><label>Availability</label><select id="dAvail"><option>Available</option><option>Busy</option></select></div>
</div>
<button class="primary" onclick="registerDonor()">Register Donor</button>
<p class="small">Prototype only. Real deployment needs secure authentication, consent and verified healthcare workflows.</p>
</div>
</section>

<section id="request" class="hidden">
  <div class="card">
    <h2>Emergency Blood Request</h2>

```

Complete source — continuation (lines 91–180)

```
<div class="formgrid">
  <div><label>Required Blood Group</label>
    <select id="rBlood"><option>A+</option><option>A-</option><option>B+</option><option>B-</option><option>AB+</option><option>AB-</option><option>O+</option><option>O-</option></select>
  </div>
  <div><label>Hospital / Request Location</label><input id="rLocation" placeholder="Area / City"></div>
  <div><label>Units Required</label><input id="rUnits" type="number" min="1" value="1"></div>
</div>
<label>Additional Details</label>
<textarea id="rDetails" placeholder="Enter necessary request details"></textarea>
<button class="primary" onclick="createRequest()">Create Request & Find Matches</button>
<div class="notice">Medical screening, donor eligibility and transfusion compatibility must be handled by qualified healthcare professionals or blood bank</div>
</section>

<section id="matches" class="hidden">
  <div class="card"><h2>Nearby Potential Matches</h2><div id="matchList"></div></div>
</section>

<section id="alerts" class="hidden">
  <div class="card">
    <h2>Donor Alerts</h2>
    <div id="alertList"><div class="match"><b>No alerts yet.</b><br>Potential match notifications will appear here after a request is created.</div></div>
  </div>
</section>

<section id="admin" class="hidden">
  <div class="card">
    <h2>Admin Dashboard</h2>
    <div class="grid">
      <div><div class="stat" id="adminDonors">0</div><p>Donors</p></div>
      <div><div class="stat" id="adminRequests">0</div><p>Requests</p></div>
      <div><div class="stat">Secure</div><p>Access-control concept</p></div>
    </div>
    <h3>Donor Records</h3>
    <table>
      <thead><tr><th>Name</th><th>Blood</th><th>Location</th><th>Status</th></tr></thead>
      <tbody id="donorTable"></tbody>
    </table>
  </div>
</section>
</div>

<footer>Community Blood Donor Finder – educational prototype, not a medical decision system.</footer>

<script>
let donors=[
  {name:"Arun",blood:"O+",location:"Coimbatore",avail:"Available"},
  {name:"Rahul",blood:"O+",location:"Coimbatore",avail:"Busy"},
  {name:"Karthik",blood:"A+",location:"Pollachi",avail:"Available"}
];
let requests=1;

function show(id){
  document.querySelectorAll("section").forEach(s=>s.classList.add("hidden"));
  document.getElementById(id).classList.remove("hidden");
  if(id==="matches"||id==="admin") render();
}

function registerDonor(){
  const name=document.getElementById("dName").value.trim();
  if(!name){alert("Please enter donor name.");return;}
  donors.push({
    name:name,
    blood:document.getElementById("dBlood").value,
    location:document.getElementById("dLocation").value.trim()||"Not specified",
    avail:document.getElementById("dAvail").value
  });
  alert("Donor registered successfully in this prototype.");
  document.getElementById("dName").value="";
  document.getElementById("dLocation").value="";
  render(); show("admin");
}

function createRequest(){
  const blood=document.getElementById("rBlood").value;
  const loc=document.getElementById("rLocation").value.trim()||"Not specified";
  const units=Math.max(1,Number(document.getElementById("rUnits").value)||1);
  const matches=donors.filter(d=>d.blood===blood && d.avail==="Available");
  requests++;

  document.getElementById("matchList").innerHTML = matches.length
    ? matches.map(d=><div class="match"><b>Alert simulated for ${escapeHtml(d.name)}</b> – ${escapeHtml(d.blood)}<br>Location: ${escapeHtml(d.location)}<br><span class="status">${d.avail}</span></div>
    : "<div class='notice'>No currently available potential donor found for this blood group.</div>";

  document.getElementById("alertList").innerHTML = matches.length
    ? matches.map(d=><div class="match"><b>Alert simulated for ${escapeHtml(d.name)}</b><br>Potential ${escapeHtml(blood)} request near ${escapeHtml(d.location)}<br><span class="status">${d.avail}</span></div>
    : "<div class='notice'>No donor alerts generated.</div>";

  render(); show("matches");
}
```

Complete source — continuation (lines 181–199)

```
function escapeHtml(value){
  return String(value).replace(/&<>"/g,ch=>({"&":"&amp;","<":"&lt;",">":"&gt;","\"":"&quot;","'":"&#39;"}[ch]));
}

function render(){
  document.getElementById("donorCount").textContent=donors.length;
  document.getElementById("requestCount").textContent=requests;
  document.getElementById("adminDonors").textContent=donors.length;
  document.getElementById("adminRequests").textContent=requests;
  document.getElementById("donorTable").innerHTML=donors.map(d=>
    `<tr><td>${escapeHtml(d.name)}</td><td>${escapeHtml(d.blood)}</td><td>${escapeHtml(d.location)}</td><td><span class="status ${d.avail==="Available"?
  `</td></tr>`).join("");
}

render();
</script>
</body>
</html>
```

Validation / limitations

This document provides a fully self-contained front-end prototype: there is no backend and data is held in JavaScript memory, so refreshing the page resets the demo state. A production system would require secure authentication, consent, persistent storage, access control, verified donor/healthcare workflows, privacy protection, and professional medical/blood-bank validation.

Source basis

The uploaded material explicitly describes donor registration, emergency blood request, blood-group and availability matching, approximate-location display, donor-alert simulation, admin donor records, a responsive single-page interface, and local browser execution without a backend. ■filecite■turn0file0■L4-L17■

Deliverables

- 1) Working HTML file: community_blood_donor_finder_working_prototype.html
- 2) Working-model PDF: this document
- 3) Test plan covering the main prototype functions.